MOTECH AUDIT

SMART CONTRACT SECURITY AUDIT

SECURE TODAY
BUILD TOMORROW

SMART CONTRACT

SECURITY AUDIT REPORT

PROJECT NAME

TREASURY VAULT AUDIT REPORT

CONTRACT FILE	TreasuryVault.sol
BLOCKCHAIN	BNB CHAIN
AUDIT DATE	19 OCT 2026
REPORT VERSION	v1.0

MOTECH AUDIT
Professional Smart Contract Security Audit Service
www.motechaudit.com

SECURITY ASSESMENT

SUMMARY

This report has been prepared for TreasuryVault smart contracts, to discover issues and vulnerabilities in the source code of their Smart Contract as well as any contract dependencies that were not part of an officially recognise ed library. A comprehensive examination has been performed, utilising Static Analysis and ManualReview techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.

The security assessment resulted in findings that ranged from critical to informational. We recommend addressing these findings to ensure a high level of security standards and industry practices. We suggest recommendations that could better serve the project from the security perspective:

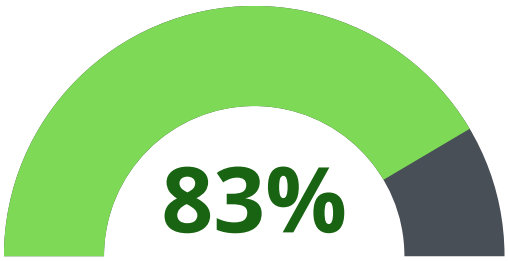
- Enhance general coding practices for better structures of source codes;
- Add enough unit tests to cover the possible use cases given they are currently missing in the repository;
- Provide more comments per each function for readability, especially contracts are verified in public;
- Provide more transparency on privileged activities once the protocol is live.

SECURITY ASSESMENT

SUMMARY

Pass/Fail	Analyzer	Description
✓	Motech Audit	Vulnerability analyzer built by Motech Audit that quickly searches within smart contracts for vulnerable code fragments not only exact but also syntactically different clones.
⚠	Formal Verifier	Sound formal verifier built by Motech Audit that analyzes within smart contracts for proving that a program satisfies nonexistence of vulnerability such as integer overflow.
✗	Achilles	Symbolic analyzer built by Motech Audit that extracts code patterns within smart contracts for finding syntactic and semantic bugs and vulnerabilities based on SWC specification.

✓ Pass ⚠ Compile Error ✗ Failed



File : TreasuryVault.sol
Language : solidity
Size. : 3792 bytes
Date : 2026-10-19T11:06:31.714Z

(STATEMENT : MOTECH AUDIT ONLY ISSUES THIS REPORT BASED ON THE FACT THAT HAS OCCURRED OR EXISTED BEFORE THE REPORT IS ISSUED, AND BEARS THE CORRESPONDING RESPONSIBILITY IN THIS REGARD. FOR THE FACTS OCCUR OR EXIST LATER AFTER THE REPORT, MOTECH AUDIT CANNOT JUDGE THE SECURITY STATUS OF ITS SMART CONTRACT. MOTECH AUDIT IS NOT RESPONSIBLE FOR IT. THE SECURITY AUDIT ANALYSIS AND OTHER CONTENTS OF THIS REPORT ARE BASED ON THE DOCUMENTS AND MATERIALS PROVIDED BY THE INFORMATION PROVIDER TO MOTECH AUDIT AS OF THE DATE OF THIS REPORT (REFERRED TO AS "THE PROVIDED INFORMATION"). MOTECH AUDIT ASSUMES THAT: THERE HAS BEEN NO INFORMATION MISSING, TAMPERED, DELETED, OR CONCEALED. IF THE INFORMATION PROVIDED HAS BEEN MISSED, MODIFIED, DELETED, CONCEALED OR REFLECTED AND IS INCONSISTENT WITH THE ACTUAL SITUATION, MOTECH AUDIT WILL NOT BEAR ANY RESPONSIBILITY FOR THE RESULTING LOSS AND ADVERSE EFFECTS. MOTECH AUDIT WILL NOT BEAR ANY RESPONSIBILITY FOR THE BACKGROUND OR OTHER CIRCUMSTANCES OF THE PROJECT.)



SECURITY ASSESMENT

DISCLAIMER

This is a limited report on our findings based on our analysis, in accordance with good industry practice as at the date of this report, in relation to cybersecurity vulnerabilities and issues in the framework and algorithms based on smart contracts, the details of which are set out in this report. In order to get a full view of our analysis, it is crucial for you to read the full report. While we have done our best in conducting our analysis and producing this report, it is important to note that you should not rely on this report and cannot claim against us on the basis of what it says or doesn't say, or how we produced it, and it is important for you to conduct your own independent investigations before making any decisions. We go into more detail on this in the below disclaimer below – please make sure to read it in full.

DISCLAIMER: By reading this report or any part of it, you agree to the terms of this disclaimer. If you do not agree to the terms, then please immediately cease reading this report, and delete and destroy any and all copies of this report downloaded and/or printed by you.

This report is provided for information purposes only and on a non-reliance basis, and does not constitute investment advice. No one shall have any right to rely on the report or its contents, and MotechAudit and its affiliates (including holding companies, shareholders, subsidiaries, employees, directors, officers and other representatives) (MotechAudit) owe no duty of care towards you or any other person, nor does MotechAudit make any warranty or representation to any person on the accuracy or completeness of the report. The report is provided "as is", without any conditions, warranties or other terms of any kind except as set out in this disclaimer, and MotechAudit hereby excludes all representations, warranties, conditions and other terms (including, without limitation, the warranties implied by law of satisfactory quality, fitness for purpose and the use of reasonable care and skill) which, but for this clause, might have effect in relation to the report. Except and only to the extent that it is prohibited by law, MotechAudit hereby excludes all liability and responsibility, and neither you nor any other person shall have any claim against MotechAudit, for any amount or kind of loss or damage that may result to you or any other person (including without limitation, any direct, indirect, special, punitive, consequential or pure economic loss or damages, or any loss of income, profits, goodwill, data, contracts, use of money, or business interruption, and whether in delict, tort (including without limitation negligence), contract, breach of statutory duty, misrepresentation (whether innocent or negligent) or otherwise under any claim of any nature whatsoever in any jurisdiction) in any way arising from or connected with this report and the use, inability to use or the results of use of this report, and any reliance on this report.

SECURITY ASSESMENT

DISCLAIMER

The analysis of the security is purely based on the smart contracts alone. No applications or operations were reviewed for security. No product code has been reviewed.

Although the assessment includes both automated security analysis and manual review, no audit can guarantee the identification of all existing or potential vulnerabilities. This report reflects the findings based on the specific smart contract code reviewed by MotechAudit at the time of the assessment. Any subsequent modifications, upgrades, or changes to the smart contracts or their dependencies are not covered by this report unless expressly stated otherwise. The project team remains responsible for addressing the findings identified in this report and validating the effectiveness of any remediation. Furthermore, the security of smart contracts may be affected by changes in the blockchain environment, external protocols, third-party dependencies, administrative privileges, and newly discovered attack techniques. Accordingly, this report should not be interpreted as a guarantee that the audited smart contracts are entirely free from vulnerabilities or will remain secure against future exploits or security incidents.

SECURITY ASSESMENT

BACKGROUND

Motech Audit was commissioned by D³ Finance to perform an audit of smart contracts:

D3Token.sol
4bcdfc70a4e408271847f12f315b81446ccad52ad9aa7d1d3201a73992fcf7c2

The purpose of the audit was to achieve the following:

- Ensure that the smart contract functions as intended.
- Identify potential security issues with the smart contract.

The information in this report should be used to understand the risk exposure of the smart contract, and as a guide to improve the security posture of the smart contract by remediating the issues that were identified.

SECURITY ASSESMENT

VULNERABILITY & RISK LEVEL

2 Total Issues

Risk represents the probability that a certain source-threat will exploit vulnerability, and the impact of that event on the organization or system. Risk Level is computed based on CVSS version 3.0.

Level	Total Issues	Vulnerability	Risk (Required Action)
Critical	0	A vulnerability that can disrupt the contract functioning in a number of scenarios,or creates a risk that the contract may be broken	Immediate action to reduce risk level.
High	0	A vulnerability that affects the desired outcome when using a contract, or provides the opportunity to use a contract in an unintended way.	Implementation of corrective actions as soon as possible.
Medium	0	A vulnerability that could affect the desired outcome of executing the contract in a specific scenario.	Implementation of corrective actions in a certain period.
Low	0	A vulnerability that does not have a significant impact on possible scenarios for the use of the contract and is probably subjective.	Implementation of certain corrective actions or accepting the risk.
Note	2	A vulnerability that have informational character but is not affecting any of the code.	An observation that does not determine a level of risk.

MOTECH AUDIT
SMART CONTRACT SECURITY AUDIT

SECURITY ASSESMENT

AUDITING STRATEGY AND TECHNIQUES APPLIED

Throughout the review process, care was taken to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices. To do so, reviewed line-by-line by our team of expert pentesters and smart contract developers, documenting any issues as there were discovered.

Methodology

The auditing process follows a routine series of steps:

1. Code review that includes the following:
- Review of the specifications, sources, and instructions provided to Motech Audit to make sure we understand the size, scope, and functionality of the smart contract.
- Manual review of code, which is the process of reading source code line-byline in an attempt to identify potential vulnerabilities.
- Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Motech Audit describe.

2. Testing and automated analysis that includes the following:
- Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
- Symbolic execution, which is analysing a program to determine what inputs causes each part of a program to execute.

3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarify, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.

4. Specific, itemized, actionable recommendations to help you take steps to secure your smart contracts.

SECURITY ASSESMENT
FINDINGS



Audit Score Summary

Category	Score (Out of 10)
Code Correctness	10
Security	9.5
Upgrade Safety	9
Access Control	10
Gas Efficiency	9
Documentation	8.5
Test Coverage	N/A
Overall Audit Score	83/100

SECURITY ASSESMENT

PROJECT OVERVIEW

TreasuryVault is a Solidity-based treasury management smart contract designed to receive ERC-20 tokens from associated contracts, such as a StakingVault, and transfer treasury funds to a designated Guardian contract. The contract is intended for deployment on BNB Chain and is implemented using Solidity version ^0.8.20.

The contract incorporates role-based access control through separate Owner and Admin roles. The Owner is authorized to update the token address, Guardian address, and Admin address, as well as transfer contract ownership. The Admin is authorized to execute treasury transfers to the configured Guardian contract. The implementation also includes a reentrancy guard and a safe-transfer helper intended to support ERC-20 tokens that return no value from transfer operations.

The contract has a relatively small codebase and a straightforward operational design. However, its security depends significantly on the trustworthiness of privileged accounts, the correctness of the configured token and Guardian contracts, and the behavior of external ERC-20 implementations. In particular, the Owner retains substantial configuration control, while the Admin can initiate transfers of available treasury tokens to the Guardian without an on-chain spending limit.

Based on the source code provided, this assessment identifies areas requiring further validation and security hardening before production deployment. The review is limited to the supplied TreasuryVault source file and does not establish the security of associated contracts, deployed addresses, or the wider project ecosystem.

VULNERABILITY FINDINGS

The entries below reproduce scanner results and provide general remediation guidance. Recommendations must be validated against the complete contract and its intended behavior.

FINDING 01 | solc-version

Informational

Detector: solc-version

Code location: Line 2

Scanner description: Version constraint ^0.8.20 contains known severe issues (https://solidity.readthedocs.io/en/latest/bugs.html) - VerbatimInvalidDeduplication - FullInlinerNonExpressionSplitArgumentEvaluationOrder - MissingSideEffectsOnSelectorAccess. It is used by: - ^0.8.20 (uploads/8b23a8c8f6e3422a987a9628ac206e61_TreasuryVault.sol#2)

Recommended action: Verify the compiler version and applicable compiler bug conditions. Select a patched compiler version compatible with the project, rebuild the contract, and re-run the checks.

FINDING 02 | low-level-calls

Informational

Detector: low-level-calls

Code location: Line 12, 13, 14, 15, 16, 17

Scanner description: Low level call in TreasuryVaultTransferHelper.safeTransfer(address,address,uint256) (uploads/8b23a8c8f6e3422a987a9628ac206e61_TreasuryVault.sol#12-17): - (success,data) = token.call(abi.encodeWithSelector(IERC20TreasuryVault.transfer.selector,to,amount)) (uploads/8b23a8c8f6e3422a987a9628ac206e61_TreasuryVault.sol#13-15)

Recommended action: Review the low-level call's return value, recipient behavior, failure handling, and surrounding state changes. A low-level call alone does not establish that a vulnerability exists.

SECURITY ASSESMENT

CONCLUSION

Based on the static review of the provided TreasuryVault.sol source code, the contract implements a straightforward treasury transfer mechanism with basic access control, balance validation, safe-transfer handling, and reentrancy protection.

No obvious classic reentrancy vulnerability was identified in the reviewed transfer flow. However, this finding should not be interpreted as confirmation that the contract is free from vulnerabilities. The review identified important trust and operational risks associated with privileged access, unrestricted treasury transfer amounts, configurable external contract addresses, and the absence of additional safeguards for high-value treasury operations.

The Owner can replace the configured token, Guardian, and Admin addresses. Meanwhile, the Admin can transfer any available balance of the configured token to the Guardian, provided the transfer succeeds. These capabilities may be consistent with the intended design, but they create significant security dependencies on the integrity of privileged accounts and the correctness of the destination contract.

Before production deployment on BNB Chain, it is recommended that the project implement stronger administrative controls where appropriate, consider multisignature governance and a two-step ownership transfer process, validate the behavior of supported tokens, and conduct integration testing against the actual Guardian and related contracts. The project should also verify compiler compatibility, deployment parameters, and the expected handling of non-standard ERC-20 tokens.

SECURITY ASSESMENT

CONCLUSION

TreasuryVault demonstrates a relatively simple implementation with several baseline security measures, but additional hardening and validation are recommended before production use. This preliminary assessment does not constitute a full ecosystem audit or a guarantee of security. A final security determination should follow testing of the complete contract system, including its dependencies and deployed configuration.

Motech Audit note: Please check the disclaimer above and note, the audit makes no statements or warranties on business model, investment attractiveness or code sustainability. The report is provided for the only contract mentioned in the report and does not include any other potential contracts deployed by Owner.